

SKILL – 8:

PART 1 — Variable Expansion:

```
GNU nano 7.2 skill8_variable.c
#include <stdio.h>
#include <string.h>
#include <stdlib.h>

#define MAX_VARS 20
#define MAX_LEN 200

struct Variable {
    char name[50];
    char value[MAX_LEN];
};

struct Variable vars[MAX_VARS];
int var_count = 0;

/* Find variable */
int find_variable(char *name)
{
    for (int i = 0; i < var_count; i++)
    {
        if (strcmp(vars[i].name, name) == 0)
            return i;
    }

    return -1;
}
```

Output:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-08$ ls
skill8_variable  skill8_variable.c
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-08$ ./skill8_variable
=====
SKILL 8 - VARIABLE EXPANSION
=====

Setting variables...

Current Variables:
NAME = Bhardwaj
CITY = Hyderabad
GREETING = Hello $NAME
MESSAGE = $GREETING from $CITY

--- Expansion Tests ---
Input : $NAME
Output: Bhardwaj

Input : $GREETING
Output: Hello Bhardwaj

Input : $MESSAGE
Output: Hello Bhardwaj from Hyderabad

Input : Welcome $NAME to $CITY
Output: Welcome Bhardwaj to Hyderabad

Input : Value = $UNKNOWN
Output: Value = [UNDEFINED]
```

PART 2 — Built-in Command Dispatch:

Code:

```
GNU nano 7.2 skill8_dispatch.c
#include <stdio.h>
#include <string.h>
#include <unistd.h>
#include <stdlib.h>

#define MAX_INPUT 200

/* Built-in command functions */
void cmd_pwd(char *args)
{
    char path[200];

    if (getcwd(path, sizeof(path)) != NULL)
        printf("Current Directory: %s\n", path);
}

void cmd_echo(char *args)
{
    if (args != NULL)
        printf("%s\n", args);
}

void cmd_set(char *args)
{
    static char state[100] = "default";
}
```

Output:

```
bhardwaj@Franathi:~/2520090182_OSSP/2520090182_Skill/Skill-08$ ./skill8_dispatch
=====
SKILL 8 - COMMAND DISPATCH
=====

Built-in commands loaded: 5

Testing dispatch logic...

> help

Available Built-in Commands:
pwd      - Display current directory
echo     - Display a message
set      - Set or display shell state
help     - Display available commands
exit     - Exit the shell

> pwd
Current Directory: /home/bhardwaj/2520090182_OSSP/2520090182_Skill/Skill-08

> echo Hello OS
Hello OS

> set running
State updated to: running

> set
Current state: running
```